

①

THE LANGUAGE OF ANALYTICS

A RESEARCH-BASED GUIDED ASSIGNMENT.

SECTION 01: FOUNDATIONAL CONCEPTS OF ANALYTICS

1. BUSINESS ANALYTICS - It is a process of identifying a business' need and recommending solutions to bring value.

Example: Introducing an online ordering system.

Problem: Customers stand for long times in queues

Investigation of the problem by a business analyst:

ANALYSING, OBSERVING, INTERVIEWING, Identifying the cause of standing in long queues, Identifying Customer Needs, Defining Requirements, Comparing solutions, Recommending one based on cost, Customer needs, and business value.

→ Focuses on what does the business need, how to improve

2. Data Analytics - It is a process of examining, cleaning, transforming, and interpreting data to identify patterns, trends and insights that can support informed decision-making.

Example: A retail store needs to better manage stock

Problem: Stock level issues across their branches.

Solution process to the problem: During analysis, the retail store realises that certain products have higher sales during festive season.

The store then use this information to make provision ~~from~~ to order more stock of that product during the peak time season.

→ Examines data to find patterns, trends, and insights

→ What does the data tell us, not what does the business need

2.

3. BUSINESS INTELLIGENCE (BI): It involves the use of technologies, processes and tools, to collect, analyse, and present business data in a way that helps the organisations make informed decisions.

BI commonly use dashboards, reports, charts, and other visualisations to turn raw data into useful business insights.

Example: A retail store can use its Business Intelligence to combine its sales and data inventory data into a Power BI dashboard. This will help stakeholders see which products sell quickly, which stores have low stocks, and how sales are changing over time. They can use this insight to manage inventory and plan stock replenishing.

→ Focus: What is happening in the business and how do we monitor it

4. Data-Driven Decision Making - It is the process of using data, facts, and analysis to guide business decisions rather than relying mainly on intuition or guess work. It involves collecting, analysing, and interpreting relevant data so that decisions can be aligned with business objectives

Example: A retail store analyses both its historical sales data and discovers that other products sell more during December. Based on this information, the company decides to increase stock of those products before the festive season.

③

5. Big Data - It refers to extremely large and complex data sets that are difficult to store, manage, and analyse using traditional data-processing systems. Big data is normally characterised by Volume, Velocity, Variety, and Variability.

Example: A large store such as Sainsbury's, generates large amounts of data from customer's transactions, online purchases, inventory systems and other sources. This data can be analysed to understand customer buying patterns, forecast demand, and manage stock.

6. Descriptive Analytics - It is the process of analysing historical data to know/understand what happened and be able to describe what happened. It summarises data using measures, reports, charts, and dashboards to identify trends and patterns.

→ Focus: What happened.

Example: A retail store analyses its sales data from the previous year and creates a dashboard showing monthly sales, total revenue and the best selling products. December information for the company is displayed on the dashboard showing what happened. This is descriptive analysis because it shows what happened.

(14)

7. Diagnostic Analytics - It is the process of analysing historical data to determine why something happened, by examining patterns, relationships and possible root causes. Some techniques include drill down analysis, correlation analysis, data mining and statistical analysis.

→ Focus: Why did it happen. [Not what happened that was discussed in descriptive analytics].

Example: A retail store notices that sales dropped significantly in one month. Using diagnostic analytics, the analyst examines sales by product, store, region, and customer segment and discovers that the decline was due to several popular products being out of stock.

8. Predictive Analytics - It is the use of historical and current data, statistical methods, and machine-learning techniques to predict likely future outcomes, trends, or behaviours.

→ Focus: What is likely to happen.

Example: A company can analyse previously sales data, seasonal trends and customer purchasing patterns to predict which products are likely to be in high demand next month. The business can therefore plan in advance.

④

9. Prescriptive Analytics - It is the process of using data, predictive models and optimisation techniques to recommend what action should be taken to achieve a desired outcome. It goes beyond ~~the~~ predicting what might happen~~ed~~ by suggesting the most appropriate course of action.

→ Focus - What should we do.

Example - A retail company uses its ^{relevant} data to prescribe which stocks need to be replenished, where, quantities in order to meet demand.

10. Data Literacy - The ability to read, understand, interpret, use, and communicate data effectively so that informed decisions can be made. It includes understanding where data comes from, interpreting charts, and graphs, recognising patterns, questioning the reliability of information, and using data appropriately.

→ Being able to communicate and understand what the data is telling you.

Example - Looking at a sales dashboard, one should be able to see and compare previous figures and interpret why there are declines in sales. Not only see that there is a decline in sales for the month.

Q

SECTION 2.

Data, Measurement and Reporting

11. Structured Data - It is data organised in a predefined format, usually rows and columns with clearly defined fields. It is commonly stored in spreadsheets and relational databases, which makes it easy to search, sort and analyse.

Example: A supermarket's sales DB ~~now~~ contains: Each piece of information has a defined field and follows structure.

Table: Products

Product-ID	Product	Date	Quantity-Sold	Sales
101	Bread	07/07/2026	50	R1000,00
102	Milk	07/07/2026	35	R700,00

Metric: Total sales = R1700

Metric: Quantity sold = 85 units.

KPI: Monthly sales target achievement = 95%

KPI: Stock availability = 98%

12. Unstructured Data - It is data that does not follow a predefined structure, format, or data model. Unlike structured data, it does not naturally fit into rows and columns in traditional relational database.

Examples: Emails, Social media posts, Customer reviews, photographs, videos, audio recordings, and PDF's.

The company can use NLP to analyse the reviews and determine whether customers have positive, negative or neutral opinions about its products.

⑥

13. Data Warehouse - It is a centralised repository that collects, cleans, integrates and stores data from multiple sources for reporting, analysis and business intelligence. Unlike an operational DB which is mainly used process day-to-day transactions a data warehouse is designed to analyse large amounts of historical.

Example: A retail company may collect ^{sales} data from different stores, its online shop, inventory system and customer database. The company can bring this information together in data warehouse. Analysts can then use SQL, Power BI or Tableau to analyse sales trends, customer behaviour and stocks levels.

The company could use its data warehouse to calculate KPI of the monthly sales growth, while also monitoring data refresh time to ensure that the dashboard receives updated information.

14. Data Mining - Is the process of using statistical methods, machine learning and other analytical techniques to discover patterns, relationships and useful information from large datasets. It can be used to identify trends, classify information, detect unusual activity and make predictions.

Example: A retail company can use data mining to analyse thousands of customer transactions and discover that customers who buy bread and milk frequently also buy eggs. This information can be used to improve product placement, promotions, or recommendations.

⑦

15. Data Visualisation - It is a process of presenting data and information visually using charts, graphs, plots, dashboards, maps, and other graphical formats. It makes complex data easier to understand and helps users identify patterns, trends, relationships and outliers.

Example - A line chart can be used to display monthly sales data of a company for the past year. Stakeholders will be able to quickly see which months had high sales and which months had low sales.

16. Dashboard - It is a visual display that brings together important data metrics, KPIs, charts, and other visualisations in one place. It provides an overview of business performance and allows users to monitor trends, identify problems, and make informed decisions.

Example - A dashboard may show that sales are currently sitting at 80% of the monthly target. This information allows the manager to investigate problem areas and focus on them to improve performance.

~~Dashboards are different from visualisations because dashboards can present different charts with different focus on the information displayed (overview). The visualisation normally displays the type of chart with info.~~

⑧

17. Metric - It is a measurable value used to track or monitor a specific aspect of business performance or activity. They help a business understand what is happening in a business process.

Example - A Supermarket wants to increase monthly Sales by 10%

They will take the following metrics into consideration

✓ Nr of products sold ✓ Nr of customer complaints

✓ Total Sales revenue

✓ Nr of customers

✓ average delivery time.

* What you measure.

18. KPI's - It is a measurable value that shows how effectively an organisation is achieving a specific business goal or objective. Unlike a Metric, A KPI is directly connected ^{linked} to an important business target.

Example: A Supermarket wants to increase monthly sales by 10%

Metric: Total monthly sales = R500,000,000

KPI: Monthly sales growth = 10% target.

Actual Results: Sales increased by 7%

The KPI therefore shows that the supermarket is 3% points below target.

Example of both: Nr of customer complaints is a metric

Reduction of number of complaints by 50% is an KPI.

* What you measure against a business goal

9.

19. Bench mark - A standard reference point against which an organisation can compare its performance. Bench marks can come from industry standards, competitors, and recognised high-performing organisations.

Example - A supermarket measures its average customer waiting time.

→ Super market's current average: 8 minutes.

→ Industry bench mark: 5 minutes

→ Difference: 3 minutes slower than the

Bench mark. The supermarket can therefore use this comparison to investigate why customers are waiting longer and identify ways to improve.

Bench mark is different from metric and KPI, bench mark focuses on the reference point you are comparing your performance against.

20. Segmentation - It is a process of dividing a large group of customers, data, or observations into smaller groups that share similar characteristics, behaviours, or needs. It allows business to analyse groups separately and make more targeted business decisions.

Example:
Frequent Shoppers - Weekly
Occasional Shoppers - Monthly
High-Value customers - > R5000 monthly
Discount-focused customers - Buys promotional ~~merch~~ products.

The Supermarket can use the data to create promotions or marketing strategies for each group rather than treating all customers the same.

10 SECTION 3: CUSTOMER & SALES KPI'S

21. Conversion rate — The percentage of people who complete a desired action after interacting with a business, website or advertisement.

Formula:
$$\text{Conversion rate} = \left(\frac{\text{Number of conversions}}{\text{Total number of interactions}} \right) \times 100.$$

Example: A clothing store receives 1000 interactions on their website, 50 of the 1000 actually buys. The conversion rate will be 5%.

22. Customer Acquisition Cost (CAC): It is the average amount of money a business spends to acquire one new customer.

Formula -
$$\text{CAC} = \frac{\text{Total sales \& marketing cost}}{\text{Number of new customer}}$$

Example - A clothing store spends R20 000 on advertising and sales during one month and gains 100 new customers.

$$\begin{aligned} \text{CAC} &= 20000 \div 100 \\ &= \text{R200} \end{aligned}$$

The business spent an average of R200 to acquire each new customer.

11

23. Customer life time value (CLV/LTV) - It is the estimated value or revenue that a business expects to receive from a customer over the entire period of their relationship with the business.

Formula $CLV = \text{Average purchase value} \times \text{Purchase frequency} \times \text{Average customer life span}$

Example A store has a customer who:
✓ Spends R500 per purchase
✓ Purchases 4 times a year
✓ Remains a customer for 3 years

$$CLV = 500 \times 4 \times 3 \\ = R6000$$

24. Churn rate - It is the percentage of customers who stop using a company's product or service during a specific period

Formula - $\text{Churn Rate} = \left(\frac{\text{Number of customers lost}}{\text{Number of customers at start of period}} \right) \times 100$

Example - starting number of customers = 1000

Cancelled subscriptions = 50

$$\text{Churn rate} = \left(\frac{50}{1000} \right) \times 100 \\ = 5\%$$

25.

12. Customer retention rate ^(CRR) - The percentage of existing customers that a business keeps over a specific period of time. It measures the effectiveness of a business keeping its clients.

Formula
$$\left[\frac{\text{Customers at end of period} - \text{New customers acquired}}{\text{Customers at start of period}} \right] \times 100$$

Example: a company has 1000 customers at the beginning of the month

Customers at end of month = 1050

New customers = 100.

$$\text{CRR} = \left[\frac{(1050 - 100)}{1000} \right] \times 100$$
$$= 95\% \text{ [Retention rate].}$$

26. Net promoter Score - It is a customer experience metric used to measure how likely customers are to recommend a company's products or services to others.

Formula -
$$\text{NPS} = \% \text{ of promoters} - \% \text{ Detectors.}$$

Example - A company surveys 100 customers:

✓ 60 are promoters

✓ 25 are passives

✓ 15 are detectors

$$\text{NPS} = 60\% - 15\% = 45$$

This company is likely to be recommended 45%

(13)

27. Average Order Value (AOV) - It is a metric that measures the average amount of money a customer spends each time they place an order with a business.

Formula -
$$AOV = \frac{\text{Total Revenue}}{\text{Number of orders}}$$

Example - A store makes R50000 from 500 orders in a month

$$\begin{aligned} AOV &= \frac{50000}{500} \\ &= R \text{ ~~50~~ } 100.00 \end{aligned}$$

The store's average order value is therefore R100.00.

28. Sales Funnel - It is a model that represents the journey a customer takes from being aware of a product or service to eventually making a purchase.

Example: A store runs a FB advertisement

- 1000 see the ad → Awareness
- 200 Click on the ad → Interest
- 50 View + add items to Their cart → Consideration
- 20 Complete Their purchase → Conversion

14

SECTION 4: Marketing and Campaign Analytics

29. Marketing campaign - A planned and coordinated set of marketing activities designed to achieve a specific business goal, such as increasing sales, creating brand awareness. It has a target audience.

Examples - A store launches a summer promotion

- ✓ Goal: Increase summer clothing sales by 20%
- ✓ Target audience: Young adults
- ✓ Activities: Instagram ads, emails, discounts in store
- ✓ Duration: 4 weeks
- ✓ Measurement: Sales revenue, number of customers and conversion rate.

30. Post campaign analysis - The process of examining data and customer information after a sale has been completed to understand how well the sale performed and what happened after the purchase.

Example: An online store sells 500 summer dresses

- ✓ Sales = 100 000 revenue
- ✓ 30 dresses returned
- ✓ Customer satisfaction: 85%
- ✓ Repeat purchases: 100 customers purchased again
- ✓ Customer complaints: 20% customers complained about delivery delays.

31. Incremental ^{Revenue} Value - The additional value, revenue, sales, conversions, or other measurable benefit that occurs because of a specific action or activity, compared with what would have happened without that action.

Formula - Incremental Value = what happened because action tests what would have happened without action

Example: An online clothing store normally generates R50 000. They run an FB ad campaign, and sales increase to R65 000.

- ✓ Sales with advertising - R65 000
- ✓ Expected sales without advertising - R50 000
- ✓ The incremental ^{Revenue} Value - R15 000.

32. Return on Investment (ROI) - It is a financial metric used to measure how much profit or benefit an investment generates compared with the cost of that investment. It is usually expressed as a percentage.

Formula - $ROI = (\text{Net profit} \div \text{Cost of Investment}) \times 100$.

Example - A business spends R10 000 on a digital marketing campaign and generates R15 000 in profit from the campaign.

$$ROI = (15000 \div R10000) \times 100 = 150\%$$

(16)

33. Return on Ad spend (ROAS) - Is a marketing metric that measures how much revenue a business generates for every rand spent on advertising.

Formula - $ROAS = \frac{\text{Revenue generated from advertising}}{\text{Advertising spend}}$

ROAS can be expressed as a ratio or percentage.

Example - A clothing store spends R5000 on a FB ad campaign and generates R20000 in sales attributed to the advertisement.

$$ROAS = R20000 \div R5000 \\ = 4:1$$

34. Click through rate - It is a digital marketing metric that measures the percentage of people who click on an advertisement, link, or listing after seeing it.

~~Formula - $CTR = (\text{Number of clicks} \div \text{Number of impressions})$~~

Formula - $CTR = \left(\frac{\text{Number of clicks}}{\text{Number of impressions}} \right) \times 100$

Example - A business displays an online advertisement 1000 times and receives 40x clicks.

$$CTR = \frac{40}{1000} \times 100 \\ = 4\%$$

The ad has 4% click through rate.

(n.)

35. Bounce rate - It is a percentage of website or App sessions where users were not considered engaged. A session is considered engaged if it lasts longer than 10 seconds.

Formula -
$$\text{Bounce Rate} = \left(\frac{\text{Non-Engaged sessions}}{\text{Total Sessions}} \right) \times 100$$

Example - An online store receives 1000,00 website visits in one month.

300 visits are not considered engaged.

700 visits are engaged.

$$\begin{aligned} \text{Bounce Rate} &= \left(\frac{300}{1000} \right) \times 100 \\ &= 30\% \end{aligned}$$

30% of the visits were not engaged.

36. A/B Testing - It is a randomised experiment where ~~two~~ or ~~two~~ more versions of a website, ad, email, or other content are shown to different people to determine which version performs better against a specific goal.

Example: ✓ "Buy Now" button
"Shop Now - 20% off" button.

The option which gets more positive feedback, will be ultimately used.